

Experiment No: 01

Experiment Name: To determine the stated value of a resistor by interpreting the color code indicated on the resistor

Course Code: EEE-1209

Course Title: Electrical Circuit lab

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Experiment Name: Resistor Color Code and Measurement of Resistance.

Theory: Resistor is a passive electrical component with two electrical circuits. The main purpose of resistor is to reduce the current flow and to lower the voltage in any particular position of the circuit. It is made of copper wires which are coiled around a ceramic rod and outer part of the resistor is coated with insulating paint.

The color code is used to read the resistance of a resistor. To read the resistor, the resistor should be held such that the tolerance band is on right with is usually gold or silver color. Starting from the left, all the color code should be noted down in sequence and the color table is used to

write the designated code for each color and determine the resistance.

Resistor Color Table:

Color	Digit	Multiplier	Tolerance
Black	0	1	
Brown	1	10	$\pm 1\%$
Red	2	100	$\pm 2\%$
Orange	3	1000	
Yellow	4	10000	
Green	5	100000	$\pm 0.5\%$
Blue	6	1000000	$\pm 0.25\%$
Violet	7	10000000	$\pm 0.1\%$
Grey	8	100000000	$\pm 0.05\%$
White	9	1000000000	
Gold		0.1	$\pm 5\%$
Silver		0.01	$\pm 10\%$

Components:

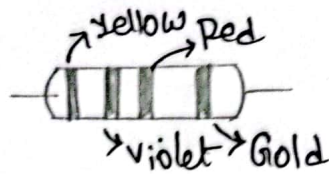
1. Set of Resistors,
2. Multimeter
3. Bread Board
4. Wire

Procedure:

1. Choose three different colour resistors.
2. Read the value by using resistor color code method for each of them.
3. Note those value down for each of the resistors.
4. Now measure the resistance using multimeter for every resistor.
5. Write down all the measured and calculated values on the table.
6. Find the deviation for the particular resistors.

Result and Discussion:

Resistor 1:

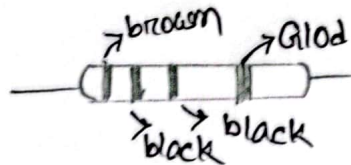


$$\text{nominal value} = 47 \times 10^2 \pm 5\% \Omega$$

$$\text{measured value} = 4630 \Omega$$

$$\begin{aligned} \text{Deviation} &= \left| \frac{4700 - 4630}{4700} \right| \times 100 \\ &= 1.48\% \end{aligned}$$

Resistor 2:

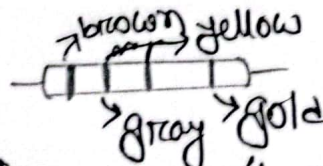


$$\text{nominal value} = 10 \times 1 \pm 5\% \Omega$$

$$\text{measured value} = 10.1 \Omega$$

$$\begin{aligned} \text{Deviation} &= \left| \frac{10 - 10.1}{10} \right| \times 100\% \\ &= \pm 1\% \end{aligned}$$

Resistor 3:



$$\text{nominal value} = 18 \times 10^4 \pm 5\% \Omega$$

$$\text{measured value} = 180500 \Omega$$

$$\begin{aligned} \text{Deviation} &= \frac{180000 - 180500}{180000} \times 100\% \\ &= 0.278\% \end{aligned}$$

Resistor	Nominal Value(R_n)	Measured value(R_m)	Deviation
R_1	4700 Ω	4630 Ω	1.48%
R_2	10.0 Ω	10.1 Ω	1%
R_3	180000 Ω	180500 Ω	0.278%

Conclusion:

As a conclusion, the resistor color code was very useful in determining the value of individual resistor selected were within the tolerance range specified by the color code. The value of resistor from digital multimeter got matched by the manual calculation using color code.